Data Integration & Quality Assessment

Steps to Generate Coverage Matrix

A San_Diego_Zips_Standardized list was defined containing every unique zip code in San Diego County. A month_range() function determined the minimum and maximum Month Year for each of the three validated datasets, which set the three-way overlap window. A scaffold of every standardized zip code × every month in that window was then checked against each dataset to build the coverage matrix, with the following columns:

- Zip Code: a 5-digit string representing each San Diego zip code
- Month Year: the month/year combination associated with each zip code
- Has_FoodBank / Has_LIHEAP / Has_CalFresh: True/False, whether that zip/month combination appears in the corresponding cleaned or validated source file
- All_Three: True/False, whether the zip/month combination appears in all three sources

Conclusions — Missingness by Data Source

The coverage matrix contains 165 unique zip codes × 7 months = 1,155 rows. The three-way overlap window is January–July 2023, set by LIHEAP's date range; CalFresh and FoodBank both cover a substantially wider range (table below).

Data Source	Min Month/Year	Max Month/Year
LIHEAP	2023-01	2023-07
CalFresh	2020-01	2025-08
FoodBank	2018-01	2025-08

- Only 58 of 165 zip codes (35.2%) appear in all three sources every month, and 76 zip codes (46.1%) never appear in all three sources in the same month. LIHEAP and FoodBank drive nearly all of the missingness; CalFresh is close to fully populated.
- LIHEAP: missing in 107 of 165 zip codes (the most of any source), driven by the source data itself only covering January–July 2023. 44 of those 107 zip codes are missing for the entire 7-month window; the remaining 63 are only partially missing (some months present, some not).
- FoodBank: missing in 63 of 165 zip codes, and unlike LIHEAP this is a strictly all-or-nothing patter, all 63 affected zip codes are missing FoodBank for the entire 7-month window, with no partial cases.

Source	Missing-Month Instances	% of All 1,155 Cells	Zips With Any Gap	Zips Missing All 7 Months
LIHEAP	577	50.0%	107 / 165	44
FoodBank	441	38.2%	63 / 165	63
CalFresh	48	4.2%	7 / 165	6

Conclusions — Missingness by City

- **Best coverage**: Lakeside and Ramona (100%), plus Valley Center, El Cajon, and Pala (72–86%) — CalFresh is nearly always present, so LIHEAP is the only gap.
- **City of San Diego** (62 zips, ~38% of the matrix) sits at 45.9% coverage, with FoodBank and LIHEAP each missing ~47% of cells — its size gives it outsized weight on the county-wide totals.
- **Rural cluster** (Dulzura, Jacumba, Guatay, Palomar Mountain, Potrero, Pine Valley, Ranchita, Tecate) shows 0% coverage from LIHEAP absence alone — likely low/no enrollment activity, not a pipeline issue.
- **La Jolla** is the one mixed-failure case: 11.4% coverage with both FoodBank (80%) and LIHEAP (86%) missing at once, unlike other low-coverage areas driven by a single source.

Zip Code Alignment Issues and Fixes

- All three datasets' zip codes were cast to string and zero-padded to five digits before joining, to prevent join misses from leading-zero loss.
- All source zip codes were confirmed to fall within the San_Diego_Zips_Standardized list; no zip codes fell outside it.

Measurement Unit Standardization Decisions

- No unit conversions were applied: each data source reports a distinct, non-comparable metric that is already internally consistent. The FoodBank dataset was reshaped from long to wide so each row returns Zip Code, Month Year, FSD_Pounds, SDFB_Pounds, and Total_Pounds_Distributed.
- The following columns were renamed for consistency across datasets:

Dataset	Old Column Name	New Column Name
LIHEAP	Outlier	LIHEAP_Outlier
LIHEAP	Enrollment_Count	LIHEAP_Enrollment_Count
CalFresh	Any_Outlier	CalFresh_Outlier
Merged	Calfresh Dollars	Calfresh_Dollars
Merged	Meal Cost	Calfresh_Meal_Cost
Merged	Calfresh Meals	Calfresh_Meals

Recommendations for Handling Gaps in Downstream Analysis

- Do not zero-fill or impute missing cells — a missing cell reflects either a data-availability gap (LIHEAP) or genuine source absence (FoodBank); zero-filling would understate real values rather than reflect true non-reporting.
- Investigate LIHEAP and FoodBank missingness as separate issues: LIHEAP (50.0% missing, 577 cells, partial gaps common) is a temporal/availability issue; FoodBank (38.2% missing, 441 cells) is strictly all-or-nothing by zip and likely needs source-side confirmation rather than a data-pull fix.
- To compare all three data sources together, filter to zip/months where all three are actually present, rather than treating missing cells as zero